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Effect of intraoral and extraoral photobiomodulation on salivary oxidative stress and inflammatory cytokines in patients with radiotherapy-induced oral mucositis

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Photobiomodulation (PBM) is an evidence-based approach for managing oral mucositis (OM) in patients undergoing head and neck radiotherapy (RT), though its biological mechanisms remain under investigation. This study evaluated salivary oxidative stress (OS) and inflammatory biomarkers in patients with head and neck squamous cell carcinoma (HNSCC) who received intraoral (IOPBM, $n = 10$) or extraoral PBM (EOPBM, $n = 8$) as supportive therapy during radiotherapy, and not as a cancer treatment. Preventive IOPBM/EOPBM were applied five times weekly from RT initiation until OM healing. Twenty healthy individuals served as the control group. Unstimulated saliva was collected at pre-, mid-, and final-RT (last RT session) to assess myeloperoxidase (MPO), malondialdehyde (MDA), superoxide dismutase (SOD), glutathione (GSH), interleukin-6 (IL-6), and interleukin-10 (IL-10). HNSCC patients exhibited altered oxidative and inflammatory biomarker profiles during RT. MDA levels showed a transient increase at mid-RT ($p < 0.05$), particularly in the IOPBM group, but were not associated with OM severity ($p > 0.05$). MPO activity was significantly higher at all RT time points and across OM severity grades compared with CT ($p < 0.05$–0.01). In contrast, SOD activity were consistently elevated throughout RT in PBM-treated patients compared with CT ($p < 0.001$), and GSH levels increased at mid-RT ($p < 0.01$). IL-6 and IL-10 levels decreased from pre- to final-RT compared to CT ($p < 0.05$). Clinically, all patients developed OM, although fewer than 40% experienced severe OM. These findings suggest that PBM enhances antioxidant defenses and modulates inflammatory responses, supporting its role in reducing RT-induced oral toxicity.

Keywords Low-level laser therapy, Oxidative stress, Inflammation, Cytokines

Oral mucositis (OM) is an inflammatory condition affecting the oral and oropharyngeal mucosa and is a frequent adverse effect of radiotherapy (RT) and/or chemotherapy (CT) in patients with head and neck squamous cell

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carcinoma (HNSCC)[1]. Severe OM can cause intense pain, impair oral function, and significantly reduce quality of life[1,2].

Oxidative stress (OS) plays a central role in the initiation and progression of OM, resulting from an imbalance between reactive oxygen species (ROS) production and antioxidant defenses[3–5]. Radiation exposure and cancer-related metabolic alterations exacerbate ROS generation, leading to molecular and cellular damage[4]. Antioxidant enzymes such as superoxide dismutase (SOD), catalase, and glutathione peroxidase (GSH) neutralize ROS, but excessive ROS production during RT can overwhelm these defenses, contributing to tissue injury and OM severity[6,7].

The damaging effects of RT are intensified by concurrent CT, which further amplifies ROS production and epithelial cell death, leading to ulceration and increased susceptibility to secondary infections[6]. Biomarkers such as SOD, GSH, myeloperoxidase (MPO), and malondialdehyde (MDA) are widely used to assess oxidative damage and inflammatory status in OM[7–11]. SOD and GSH play key roles in redox homeostasis, whereas MPO and MDA reflect inflammatory and lipid peroxidation processes, respectively[12–18].

Inflammatory cytokines are also critical mediators in OM pathogenesis. ROS-induced activation of transcription factors such as NF-κB promotes the release of pro-inflammatory cytokines, including tumor necrosis factor-alpha, interleukin-1 beta, and interleukin-6 (IL-6)[19]. IL-6 is particularly associated with symptom severity and sustained inflammation in OM[19–21]. Conversely, interleukin-10 (IL-10) exerts anti-inflammatory effects and contributes to immune homeostasis; an imbalance between IL-6 and IL-10 may exacerbate tissue damage during oncological treatment[20–22].

Photobiomodulation (PBM) has demonstrated efficacy in reducing inflammation and promoting tissue repair, although its effects depend on treatment parameters[23,24]. Low-power lasers are commonly used for anti-inflammatory and regenerative purposes, while high-power lasers, when defocused, can also induce PBM effects with deeper tissue penetration and clinical versatility, particularly for extraoral applications[25–30]. PBM may modulate ROS levels in a context-dependent manner, increasing ROS in healthy cells and reducing oxidative damage in stressed tissues[6,19]. However, the effects of PBM on OS and cytokine modulation under different clinical protocols, including defocused high-power laser applications, remain insufficiently explored.

Given the potential of PBM in regulating oxidative and inflammatory pathways, this study aimed to evaluate salivary levels of OS markers (MDA and MPO), antioxidants (SOD and GSH), and inflammatory cytokines (IL-6 and IL-10) in HNSCC patients undergoing RT and/or CT treated with intraoral or extraoral PBM protocols for OM management.

Results

Clinical and demographic data

The clinical and demographic characteristics of the participants are presented in Table 1. Among the oncology patients, ten (55.5%) were allocated to the IOPBM, and eight (44.5%) to the EOPBM, totaling 18 participants. The mean age was 59 years (± 10.1). Participants were predominantly male ($n = 16$, 88.9%), most of whom had a history of chronic exposure to risk factors such as tobacco use ($n = 15$, 83.4%) and alcohol consumption ($n = 13$, 72.2%). All cancer patients were diagnosed with HNSCC, with 11 cases (61.1%) located in the oral cavity and 7 cases (38.9%) in the oropharynx. Regarding clinical staging, advanced stages (III and IV) were the most prevalent, accounting for more than 75% of the cases. All oncology patients underwent head and neck RT, with the following treatment combinations: RT + CHT ($n = 8$, 44.5%), RT + surgery + CHT ($n = 7$, 38.9%), and RT + surgery ($n = 3$, 16.7%). The mean RT dose was 59.2 Gy (± 6.18).

Among the twenty healthy control participants, only age, sex, and salivary flow data were collected, as other potential confounding variables were excluded based on the study's inclusion and exclusion criteria (e.g., risk factors, comorbidities, oral diseases or conditions, use of anti-inflammatory or antioxidant medications). The mean age of the control group was 54 years (± 19.7), with no sex predominance (male: $n = 10$, 50%; female: $n = 10$, 50%).

Oral adverse effects resulting from head and neck RT

Findings revealed that all patients experienced some level of OM during RT, but less than 40% developed severe forms, with no significant difference between the EOPBM and IOPBM groups ($p = 0.86$ mid-RT WHO; $p = 0.27$ end-RT WHO; $p = 0.35$ mid-RT NCI; $p = 0.91$ end-RT NCI). Salivary flow declined mid-treatment but partially recovered by the end, reaching levels similar to the control group. Xerostomia and dysphagia scores remained low throughout, with no group differences. Dysgeusia worsened significantly by the end of RT ($p < 0.001$ IOPBM; $p = 0.03$ EOPBM). Oral pain scores remained low (< 4), suggesting a beneficial effect of PBM.

Transient increases in lipid peroxidation marker during RT under IOPBM and EOPBM protocols

In the comparative analysis of the PBM cohort, statistically significant differences were observed between the CT and mid-RT, as well as between pre-RT and mid-RT ($p < 0.05$) (Fig. 1a). When stratified by PBM protocol, a significant difference was observed only in the IOPBM group between CT and mid-RT ($p < 0.01$) (Fig. 1b), whereas no statistically significant differences were detected in the EOPBM group (Fig. 1c).

In the analysis of this marker in the context of OM severity at the end of RT (Fig. 1d), no significant differences in MDA levels were observed across different mucositis severity grades or when compared with the control group ($p > 0.05$).

These findings indicate a transient increase in lipid peroxidation levels during mid-RT, which may reflect radiation-induced oxidative stress. Nevertheless, MDA levels were not associated with OM severity, suggesting a limited relationship between MDA modulation and clinical mucositis outcomes in this cohort (Fig. 1a–d).

Characteristics	IOPBM ($n=10$)	EOPBM ($n=8$)	Total ($n=18$)	p-value
Patients, n (%)	10 (55.6)	8 (44.4)	18 (100)	—
Age, years, mean (SD)	61.3 (10.5)	56.1 (9.4)	59.0 (10.1)	0.03
Sex, n (%)				
Male	9 (90.0)	7 (87.5)	16 (88.9)	0.16
Female	1 (10.0)	1 (12.5)	2 (11.1)	
Risk factors, n (%)				
Tobacco use	8 (80.0)	7 (87.5)	15 (83.3)	0.17
Alcohol consumption	6 (60.0)	7 (87.5)	13 (72.2)	0.21
Tumor site, n (%)				
Oral cavity	6 (60.0)	5 (62.5)	11 (61.1)	1.0
Oropharynx	4 (40.0)	3 (37.5)	7 (38.9)	1.0
Clinical stage, n (%)				
Oral cavity ($n=11$)				
Stage II	1 (10.0)	2 (25.0)	3 (16.7)	0.52
Stage III	1 (10.0)	1 (12.5)	2 (11.1)	
Stage IV	4 (40.0)	2 (25.0)	6 (33.3)	
Oropharynx ($n=7$)				
Stage II	1 (10.0)	0 (0)	1 (5.6)	0.92
Stage III	1 (10.0)	0 (0)	1 (5.6)	
Stage IV	2 (20.0)	3 (37.5)	5 (27.8)	
Treatment, n (%)				
RT + CHT	4 (40.0)	4 (50.0)	8 (44.4)	1.0
RT + Surgery + CHT	5 (50.0)	2 (25.0)	7 (38.9)	
RT + Surgery	1 (10.0)	2 (25.0)	3 (16.7)	
RT dose, Gy, mean (SD)	62.2 (4.2)	55.5 (6.5)	59.2 (6.2)	0.08

Table 1. Demographic and clinical characteristics of the study population.

Temporal changes in salivary MPO levels during RT across PBM protocols

In response to RT-induced tissue injury, neutrophils become activated and release MPO, an enzyme widely used as a marker of immune cell–mediated oxidative stress. In the comparative analysis of the combined PBM cohort (IOPBM/EOPBM) (Fig. 1e) and the EOPBM group (Fig. 1g), MPO activity was significantly increased at pre-, mid-, and final RT compared with the CT ($p<0.05$). In the IOPBM group (Fig. 1f), significant differences were observed at all RT time points compared with the CT, as well as between pre-RT and mid-RT and between pre-RT and final RT ($p<0.05$).

In the analysis of OM severity in final RT (Fig. 1h), MPO activity was significantly higher in patients with mild and severe mucositis compared with the control group ($p<0.01$).

These findings indicate that RT was associated with increased neutrophil-related oxidative activity across PBM protocols and OM severity grades.

The antioxidant mechanism mediated by SOD is activated prior to RT

SOD is one of the primary enzymatic defense mechanisms against ROS and is regarded as a key antioxidant marker. In our analysis, SOD levels were significantly elevated and remained stable in both IOPBM and EOPBM groups before the initiation of RT, as well as during mid- and post-treatment periods ($p<0.001$), when compared to the CT (Fig. 1i).

In the analysis of OM at the end of RT (Fig. 1l), SOD levels were high in comparison of No/mild mucositis ($p<0.01$) and severe mucositis ($p<0.01$) to the CT. These findings suggest that PBM may contribute to the maintenance of elevated SOD levels, supporting its role in enhancing antioxidant defense during RT.

GSH levels increase during mid-RT under PBM protocols

GSH is part of the enzymatic defense system involved in regulating OS and it is considered a key antioxidant marker. In our analysis (IOPBM/EOPBM) and IOPBM only, GSH levels were significantly elevated when comparing the CT to mid-RT ($p<0.01$) (Fig. 1m).

The increase in GSH levels observed at mid-RT suggests that PBM exerts an antioxidant effect after the onset of RT, suggesting local effect. However, when evaluating this marker with OM severity at final RT (Fig. 1p), no statistically significant differences were observed between groups.

Pro-inflammatory cytokine IL-6 levels decrease during RT under PBM protocols

IL-6 is a pro-inflammatory cytokine produced during the immune and inflammatory response to oxidative and tissue damage induced by RT. Our analysis demonstrated a reduction in IL-6 levels when comparing the CT to

Characteristics	IOPBM ($n=10$)	EOPBM ($n=8$)	Total ($n=18$)	p-value
OM at mid-RT – WHO, n (%)				
No/Mild (Grades 0–2)	9 (90.0)	7 (87.5)	16 (88.9)	0.86
Severe (Grades 3–4)	1 (10.0)	1 (12.5)	2 (11.1)	
OM at final RT – WHO, n (%)				
No/mild (Grades 0–2)	5 (50.0)	6 (75.0)	11 (61.1)	0.27
Severe (Grades 3–4)	5 (50.0)	2 (25.0)	7 (38.9)	
OM at mid-RT – NCI, n (%)				
No/mild (Grades 0–2)	9 (90.0)	8 (100)	17 (94.4)	0.35
Severe (Grades 3–4)	1 (10.0)	0	1 (5.6)	
OM at final RT – NCI, n (%)				
No/mild (Grades 0–2)	6 (60.0)	5 (62.5)	11 (61.1)	0.91
Severe (Grades 3–4)	4 (40.0)	3 (37.5)	7 (38.9)	
Salivary flow rate, mL/min, mean (SD)				
Control group	—	—	0.76 (0.37)	—
Pre-RT (baseline)	1.04 (0.93)	1.29 (0.88)	1.15 (0.89)	0.57
Mid-RT	0.40 (0.24)	0.69 (0.45)	0.53 (0.37)	0.23
Final RT	0.85 (0.50)	0.67 (0.43)	0.77 (0.47)	0.63
Within-group p-value	0.12	0.33	0.23	—
Xerostomia score (0–3), mean (SD)				
Pre-RT (baseline)	0.50 (0.52)	0.87 (0.64)	0.66 (0.59)	0.22
Mid-RT	0.50 (0.52)	0.85 (0.37)	0.64 (0.49)	0.15
Final RT	0.80 (0.63)	0.87 (0.64)	0.83 (0.61)	0.83
Within-group p-value	0.45	1.00	0.45	—
Dysgeusia score (0–2), mean (SD)				
Pre-RT (baseline)	0.30 (0.67)	0.62 (0.91)	0.44 (0.78)	0.17
Mid-RT	1.70 (0.67)	1.00 (1.07)	1.39 (0.91)	0.14
Final RT	2.00 (0.00)	1.88 (0.35)	1.94 (0.23)	1.00
Within-group p-value	< 0.001	0.03	< 0.001	—
Dysphagia score (0–4), mean (SD)				
Pre-RT (baseline)	1.20 (0.78)	1.25 (1.04)	1.22 (0.87)	0.81
Mid-RT	1.60 (0.96)	1.38 (1.06)	1.50 (0.98)	0.71
Final RT	2.10 (0.87)	2.00 (0.75)	2.06 (0.80)	0.81
Within-group p-value	0.11	0.31	0.23	—
Pain (VAS 0–10), mean (SD)				
Pre-RT (baseline)	0.70 (1.64)	3.63 (2.77)	2.00 (2.61)	0.02
Mid-RT	1.90 (3.14)	4.00 (3.21)	2.83 (3.26)	0.21
Final RT	3.30 (4.16)	3.38 (2.88)	3.33 (3.55)	0.78
Within-group p-value	0.25	0.97	0.13	—
Mouth opening (mm), mean (SD)				
Pre-RT (baseline)	42.4 (14.3)	40.6 (16.7)	41.6 (15.0)	0.81
Mid-RT	40.6 (16.9)	41.5 (15.6)	41.0 (15.8)	0.96
Final RT	37.6 (16.5)	36.6 (17.5)	37.1 (16.6)	0.95
Within-group p-value	0.71	0.82	0.97	—
Trismus, n (%)	5 (50.0)	3 (37.5)	8 (44.4)	0.06

Table 2. Oral effects in patients undergoing RT.

the final of RT ($p<0.01$), as well as between the pre-RT and final-RT timepoints in the groups IOPBM/EOPBM and IOPBM only ($p<0.01$) (Fig. 1q and r).

These findings suggest that PBM is capable of decreasing IL-6 levels throughout RT. Additionally, in the analysis of OM at the end of RT (Fig. 1t), patients with severe mucositis exhibited significantly lower IL-6 levels compared to the CT ($p<0.001$), indicating a modulatory effect on the inflammatory response.

Anti-inflammatory cytokine IL-10 levels decrease during RT under PBM protocols

IL-10 is an anti-inflammatory cytokine that plays a key role in regulating excessive inflammation, such as that observed in mucositis. Our analysis revealed a reduction in IL-10 levels at final RT compared to the CT in all groups ($p<0.05$) (Fig. 1u-w).

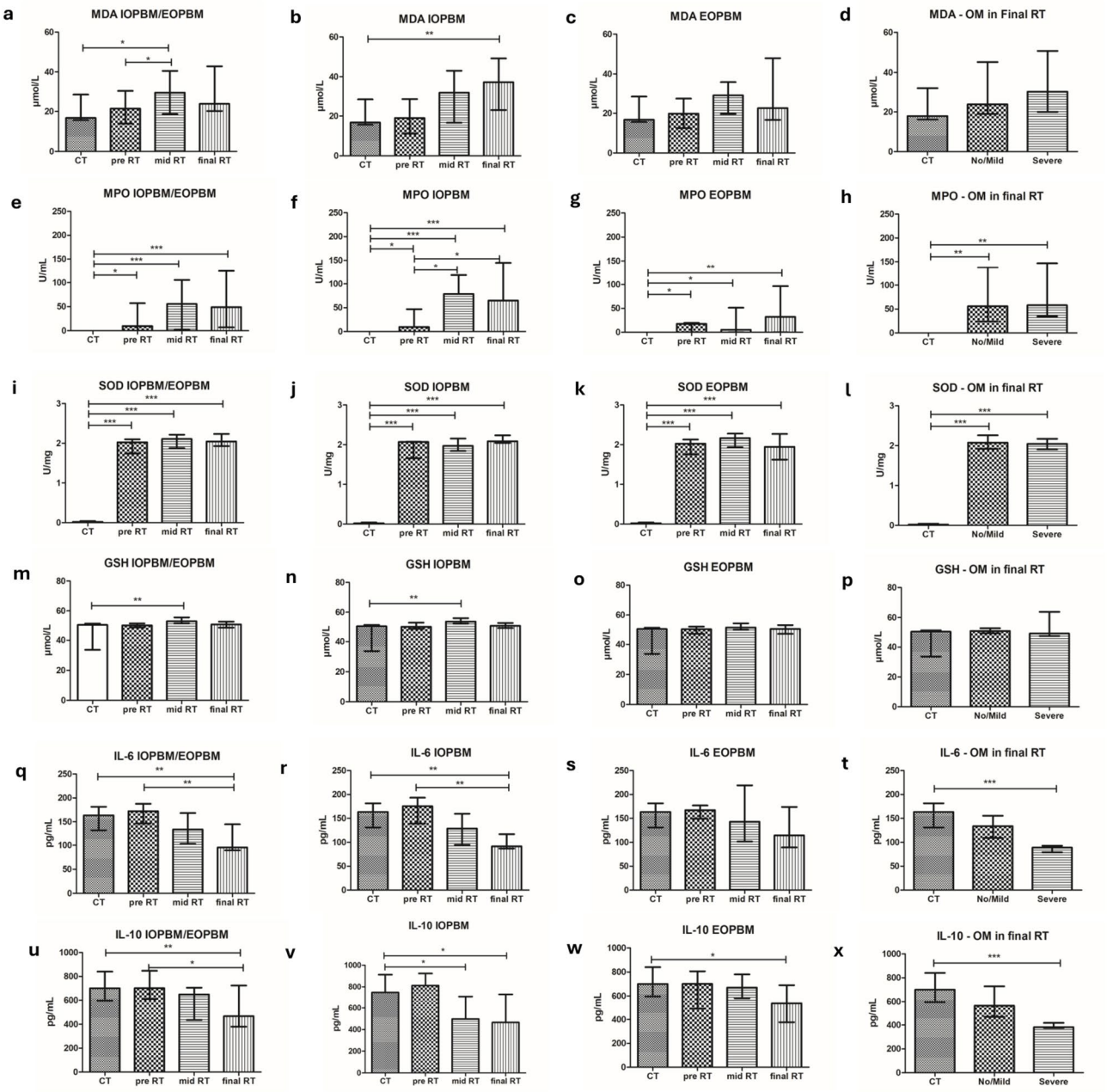


Fig. 1. Comparative evaluation of salivary levels of MDA (**a**–**d**), MPO (**e**–**h**), SOD (**i**–**l**), GSH (**m**–**p**), IL-6 (**q**–**t**), and IL-10 (**u**–**x**) in HNSCC patients undergoing RT and submitted to IOPBM or EOPBM (IOPBM/EOPBM, n = 18; IOPBM, n = 10; EOPBM, n = 8), compared with healthy controls (CT, n = 20). Data are presented as median with interquartile range (IQR), as measures of dispersion. Intragroup comparisons across time points (pre-, mid-, and final RT) were performed using the Friedman test with Dunn's post hoc test. Between-group comparisons between CT and each RT time point were performed using the Mann–Whitney U test. $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

In the evaluation of OM at the final-RT (Fig. 1x), a decrease in IL-10 levels was also observed in cases of severe mucositis when compared to the CT ($p < 0.001$). These findings indicate that IL-6 and IL-10 levels were reduced during PBM exposure, suggesting a potential modulation of inflammatory pathways.

Discussion

The primary objective of this study was to evaluate the effects of PBM in cancer patients undergoing head and neck RT, with a focus on oral adverse effects, OS markers, and inflammatory mediators. The results indicate that PBM, regardless of the protocol used (IOPBM or EOPBM), has a positive impact on pain control and the onset of OM, while also modulating OS and inflammation levels. Both PBM protocols demonstrated comparable and favorable outcomes in mitigating RT-induced adverse effects, including OM severity and antioxidant imbalance.

These findings provide mechanistic insight into PBM-associated biological responses during RT and expand current evidence on supportive care strategies in oncology.

Consistent with previous randomized clinical trials[28,30,31], both PBM protocols were associated with low OM severity and pain scores, although dysgeusia progressively worsened throughout RT. Randomized clinical studies have demonstrated that PBM can delay OM onset and reduce pain and pharmacological analgesic use in irradiated patients, supporting its clinical utility as a supportive care intervention. The present observational findings align with these clinical data, although causal inference is limited by the study design.

Recent interventional studies investigating oral-systemic supportive care interventions provide additional context for our biomarker findings. A randomized trial evaluating oral glutamine supplementation reported reduced salivary TGF-β1 levels and improved clinical outcomes, including reduced mucositis severity and pain, highlighting the potential of biomarker-guided supportive therapies in RT-induced OM[32]. Similarly, immune-enhancing nutritional interventions such as L-arginine and glutamine have demonstrated wound healing and immunomodulatory effects in head and neck cancer patients, emphasizing the relevance of systemic and local redox and inflammatory pathways in mucosal toxicity[33]. Large multicenter cohort studies have also confirmed the high prevalence of xerostomia and hyposalivation after RT, particularly in patients with advanced disease stage and concurrent CHT + RT, supporting the clinical relevance of salivary biomarker monitoring in this population[34,35].

MDA, a classical marker of lipid peroxidation, exhibited a transient increase at mid-RT, likely reflecting cumulative radiation-induced oxidative stress during the acute phase of mucosal injury. However, MDA levels were not associated with OM severity, suggesting that lipid peroxidation dynamics may not directly correlate with clinical mucositis burden under PBM exposure. These findings contrast with previous reports showing increased post-RT MDA levels in patients not receiving PBM, and may indicate that PBM exposure was associated with attenuation of lipid peroxidation progression rather than complete prevention of oxidative damage[7]. Nevertheless, given the observational nature of this study, the potential modulatory role of PBM on lipid peroxidation should be interpreted cautiously.

MPO activity was significantly elevated throughout RT and across OM severity grades, indicating persistent neutrophil activation and immune-mediated oxidative activity. The lack of detectable modulation of MPO across PBM protocols suggests that PBM exposure may not markedly influence neutrophil-driven oxidative pathways under the conditions evaluated. This observation aligns with experimental studies demonstrating that PBM primarily affects mitochondrial and antioxidant pathways rather than immune cell recruitment or activation[36]. Moreover, salivary MPO reflects both local and systemic inflammatory processes, which may limit its specificity as a mucosal tissue biomarker.

The analysis of the antioxidant markers revealed that SOD levels remained consistently elevated in patients receiving PBM at all evaluated time points—before, during, and after RT—when compared to the control group ($p < 0.001$). This persistent elevation indicates that PBM may help preserve the primary enzymatic defense system against OS, potentially enhancing resilience to RT-induced damage, even if it does not directly reduce ROS levels. Importantly, the role of SOD as a therapeutic target in OM has been increasingly emphasized. Sonis (2021)[13] highlighted that pharmacological modulation of this pathway—particularly strategies aimed at increasing SOD levels—may significantly reduce the severity of OM. Two promising approaches have been explored: gene therapy to enhance SOD expression and the use of dismutase mimetics. Among these, Avasopasem manganese (GC4419), a synthetic SOD mimetic, has advanced to phase 2 clinical trials[37], demonstrating radioprotective effects and significantly reducing the duration, severity, and incidence of severe OM. Our findings support the hypothesis that PBM may act through similar antioxidant mechanisms, offering a non-invasive alternative for enhancing endogenous defenses such as SOD, and reinforcing its relevance in OM prevention and supportive cancer care.

Still regarding the antioxidant effects of PBM, the scoping review by Nguyen et al.[38], which evaluated studies on OS and OM induced by chemoradiotherapy, demonstrated that PBM, particularly extraoral application, reduces the severity of OM. This reduction has been associated with decreased ROS production and increased activity of both SOD and GSH[38,39]. GSH levels were significantly increased at mid-RT, particularly in the IOPBM group, suggesting protocol-dependent modulation of antioxidant pathways. This finding supports the hypothesis of a local PBM effect. Differences in optical properties, penetration depth, and irradiated tissue volume between protocols may explain the preferential GSH modulation observed with IOPBM. However, GSH levels were not associated with OM severity, indicating that antioxidant upregulation may not directly translate into clinical mucositis outcomes[38,39].

IL-6 and IL-10 levels decreased during RT under PBM exposure, suggesting an association with modulation of inflammatory signaling pathways. IL-6 reduction may reflect attenuation of RT-induced inflammatory cascades, consistent with previous studies reporting decreased IL-6 following PBM[19,40–42]. Interestingly, IL-10 levels also decreased, diverging from studies reporting[19,40,41] compensatory IL-10 increases during PBM. This pattern may reflect a balanced immunoregulatory response in which inflammation is attenuated at its origin without requiring compensatory anti-inflammatory cytokine upregulation. This dual modulation profile aligns partially with Oton-Leite et al.[40], who reported a non-significant trend toward IL-10 reduction, and collectively underscores the capacity of PBM to reestablish immune homeostasis, contributing to the observed clinical benefits. Nevertheless, these findings should be interpreted cautiously given the observational design and the evaluation of a limited cytokine panel.

While PBM exposure may be associated with modulation of oxidative and inflammatory biomarkers, these changes were not consistently associated with OM severity. This biomarker–clinical discordance has also been observed in interventional supportive care trials, where biomarker modulation did not always predict clinical outcomes[32,33]. These findings highlight the complexity of RT-induced mucositis pathophysiology and suggest that salivary biomarkers may reflect biological processes that do not directly translate into clinical severity scales.

Despite the significant contributions of this study, several limitations should be acknowledged. The relatively small sample size (18 cancer patients and 20 healthy controls) may limit the generalizability of the findings and reduce statistical power for subgroup analyses. This was a single-center study, which may introduce center-specific biases related to patient characteristics, clinical management, and PBM application protocols, thereby limiting external validity. Nevertheless, the analysis across three distinct time points strengthens the study design and supports the interpretation of temporal trends. The lack of long-term follow-up also limits conclusions regarding the durability and clinical relevance of PBM-related effects. Future multicenter studies with larger cohorts and longitudinal follow-up are warranted to confirm these findings. Although randomized controlled designs would further clarify the effects of PBM on oxidative stress and inflammatory pathways, the inclusion of RT-only comparison groups raises ethical concerns given existing guideline recommendations supporting PBM for mucositis management. Multicenter cohorts with standardized PBM protocols, longitudinal biomarker profiling, multivariable modeling, and mechanistic assays will be essential to better elucidate protocol-dependent biological effects and the relationship between biomarkers and clinical outcomes.

In conclusion, PBM exposure during head and neck RT was associated with favorable clinical outcomes and possible modulation of oxidative and inflammatory biomarkers. Transient increases in lipid peroxidation and persistent neutrophil-related oxidative activity were observed, whereas antioxidant pathways, particularly SOD and GSH, showed protocol-dependent modulation. Reductions in IL-6 and IL-10 levels suggest potential immunoregulatory effects under PBM exposure. These findings provide mechanistic insight into PBM-associated biological responses and support its role as a supportive care intervention during RT, although causal inference remains limited. Further mechanistic and controlled studies are required to optimize PBM protocols and establish biomarker-guided supportive care strategies in oncology.

Methods

Ethical considerations

All procedures performed in this study were conducted in accordance with the ethical standards of the Brazilian National Research Ethics Committee, as well as the 1964 Declaration of Helsinki and its subsequent amendments, or comparable ethical standards. The study was approved by the Ethics Committee of the Hospital de Clínicas de Porto Alegre, Rio Grande do Sul, Brazil (approval number 2020 – 0189). All patients provided written informed consent.

Study design

This is an analytical cross-sectional study that evaluated saliva samples and clinical data from patients diagnosed with HNSCC, as described in the study by Klein et al.[30]. STROBE guidelines were followed and a flowchart of the study is presented (Fig. 2). These patients were treated with RT or chemoradiotherapy (RT + CHT) and received two distinct PBM protocols for the prevention and treatment of OM, with 10 patients receiving IOPBM protocol

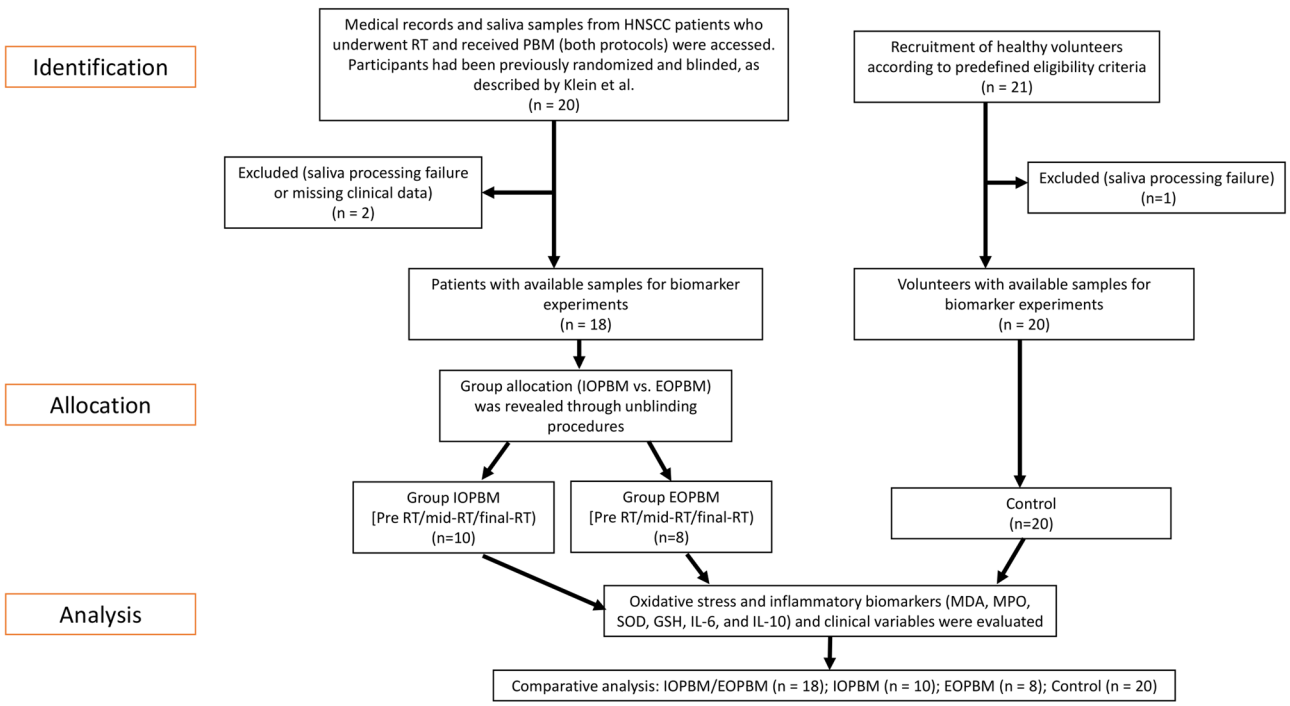


Fig. 2. Flow diagram of the study.

and 8 to EOPBM protocol. Twenty healthy volunteers, matched in age to the intervention group and without any systemic or local oral conditions, were selected as controls.

Inclusion and exclusion criteria

The inclusion criteria followed those established in the study by Klein et al.[30]. Saliva samples and clinical data were collected from patients with HNSCC, who had undergone oral health optimization to eliminate any local oral disease (such as periodontitis, dental caries, or bacterial plaque) and received RT, with or without surgery, and with or without concurrent CHT. All included patients underwent conventional fractionated RT protocols (2 Gy per day, excluding weekends) for 5 to 7 weeks, using intensity-modulated RT (IMRT). The CHT protocol consisted of cisplatin (100 mg/m^2) or cisplatin (50 mg/m^2).

Exclusion criteria included the use of any type of anti-inflammatory drugs or antioxidant supplementation, a history of chronic diseases that could affect healing, or previous treatment with RT and/or CHT.

Inclusion criteria for healthy volunteers were individuals over 18 years of age, without comorbidities, free from any active oral disease (such as periodontitis, dental caries, or bacterial plaque), and not taking continuous medication that could affect salivary flow. Exclusion criteria for healthy volunteers included: use of anti-inflammatory drugs or antioxidant supplementation, any disease that could affect healing or salivary flow, and any previous treatment for salivary gland disorders or cancer therapy.

Saliva collection and sialometry

Unstimulated saliva collection was conducted by Klein et al.[30]. Unstimulated sialometry was performed in a private setting, with the participant seated, following a minimum interval of two hours after oral hygiene or the last meal. All oncology patients were orientated to use 0.12% chlorhexidine rinse twice a day to maintain oral health. Both patients and healthy volunteers were instructed to deposit all saliva into a clean, unused 50 mL Falcon tube for five minutes. For oncology patients, saliva samples were collected at three time points: immediately before the first RT session (pre-RT), midway through treatment between the 15th and 20th session of RT (mid-RT), and on the final day of RT (final-RT). For healthy volunteers, only one saliva sample was collected. Saliva samples were collected at standardized time points, taking circadian rhythms into account. After collection, the tubes were placed in a refrigerated container. The total saliva volume was measured, and the samples were aliquoted and stored at − 80 °C until further analysis. Salivary flow rate was calculated by dividing the total volume (in mL) by five, yielding a final flow rate expressed in mL/min.

Clinical and sociodemographic data of the patients

Clinical data were obtained from patients' electronic medical records stored in the The Research Electronic Data Capture (REDCap) platform (14.0.16, Vanderbilt University, Nashville, Tennessee, USA) at the Hospital de Clínicas de Porto Alegre. The collected sociodemographic variables included age, sex, risk factors, tumor site, clinical stage, type of oncological treatment, and RT dosage.

Clinical parameters assessment

Data on adverse effects related to RT were obtained from medical records at three time points: before RT (pre-RT), between the 15th and 20th session of RT (mid-RT), and at the end of RT (final-RT). Clinical variables including oral OM, mouth opening and trismus, xerostomia, dysgeusia, dysphagia, and pain (assessed using the visual analog scale)were evaluated and categorized according to the protocols described below.

OM assessment

The severity of OM was assessed based on Klein et al.[30], using the criteria established by the World Health Organization (WHO) and the National Cancer Institute (NCI). OM was classified as severe when scored as grade 3 or 4 on the WHO and NCI scale. Clinical data regarding OM and dietary intake were collected at the same time points as the saliva samples (pre-RT, mid-RT, and final-RT) from patient records.

Mouth opening and trismus assessment

Mouth opening and trismus were evaluated by Klein et al.[30], based on the protocols by the NCI Common Terminology Criteria for Adverse Events version 4.0. Trismus was assessed using a caliper in the three time points (-pre, -mid and -final RT). The measurement of mouth opening was performed using anatomical landmarks of the maxilla and mandible, referencing either the central incisors (for dentate patients) or the alveolar ridge (for edentulous patients). Patients were seated with relaxed shoulders and their heads aligned in the midline; they were then asked to open their mouths as wide as possible for measurement. Patients with a mouth opening of less than 35 mm were classified as having trismus.

Xerostomia assessment

Xerostomia was evaluated both subjectively and objectively and categorized according to the scale described by Klein et al.[30], ranging from 0 to 3: 0 – no salivary alteration; 1 – mild dryness without habit changes; 2 – dry mouth requiring liquids to swallow; 3 – dry mouth causing changes in diet and interfering with sleep, speech, or other activities. Mean xerostomia scores were recorded at pre-RT, mid-RT, and final-RT.

Dysphagia and dysgeusia assessment

Dysphagia and dysgeusia were assessed based on adapted parameters from the study by Klein et al.[30]. Patients completed a standardized form using the categorical scoring system described below. For dysphagia, the following scale was used: 0 – no swallowing difficulties; 1 – difficulty swallowing solid foods; 2 – difficulty swallowing both solids and liquids; 3 – use of feeding tube or intravenous hydration; 4 – complete obstruction,

Parameter	IOPBM	EOPBM
Wavelength (nm)	660 ± 10 nm	810 nm + 980 nm (50%/50%)
Operating mode	Continuous	Pulsed
Frequency	~ 50/60 Hz	50 Hz
Pulse duration	Continuous	1 ms
Peak power	0.01 W	10 W
Average power	100 mW	1000 mW
Spot area	0.03 cm^2	4.91 cm^2
Beam shape	Round	Round
Beam profile	Gaussian	Gaussian
Irradiance at target	3.333 mW/cm^2	407 mW/cm^2
Fluence	10 J/cm^2	6.11 J/cm^2
Photon Fluence (Einstein)	4.2	1.9
Exposure time per point	3 s	30 s
Energy per point	0.3 J	30 J
Number of irradiated points	33	10
Application technique	Contact	Contact
Session frequency	5x/week	5x/week
Duration of the treatment	1 st session of RT until last session or OM healing	1 st session of RT until last session or OM healing

Table 3. Parameters of PBM protocols.

inability to swallow, including saliva. For dysgeusia, the scale was: 0 – no taste alteration; 1 – mild taste alteration; 2 – severe taste alteration. Mean scores for both variables were recorded at the three time-points.

Pain assessment

Oral pain was subjectively assessed using a visual analog scale (VAS) as described by Klein et al.[30], where "0" indicated no pain and "10" indicated the maximum pain. VAS-related data were collected, and mean values were reported at the three time points.

PBM parameters

Preventive IOPBM and EOPBM protocols were administered five times per week, starting on the first day of RT and continuing until the healing of ulcerated OM lesions as described by Klein et al.[30]. All PBM sessions were applied by a single trained professional, and the clinician and the patient used protective eyewear. All PBM parameters used in the study are detailed in Table 3.

IOPBM and EOPBM applications

Both PBM protocols followed the parameters described by Klein et al.[30]. Protocol assignment was randomized using an institutional application based on a sequential allocation process. IOPBM was administered using a continuous wave InGaAlP diode laser (MM Optics Ltd., São Carlos, Brazil) in contact mode, perpendicular to the oral mucosa, avoiding the tumor site. A total of 33 intraoral points were irradiated, including buccal and labial mucosa, lateral and ventral surfaces of the tongue, and floor of the mouth. EOPBM employed a pulsed diode laser (Gemini Diode Laser, Azena Medical LLC, USA) with dual wavelengths (810/980 nm), targeting 10 points across the face and anterior neck, including bilateral submandibular and submental regions (Table 3).

Oxidative stress assay

MDA levels

Malondialdehyde (MDA) levels were determined in duplicate, following the protocol established by Mármora et al.[9], with adaptations. Saliva samples were thawed on a refrigerated surface, homogenized in Tris buffer at a 1:5 (v/v) ratio, and centrifuged at 11,000 rpm for 10 min at 4 °C. Subsequently, 300 µL of the supernatant were mixed with 750 µL of the chromogenic reagent 1-methyl-2-phenylindole (Sigma, St. Louis, MO, USA) and 225 µL of HCl (37%). The mixture was incubated in a water bath at 45 °C for 40 min. After a second centrifugation under the same conditions, 300 µL of the supernatant were transferred to a 96-well plate. Absorbance was measured at 586 nm using a spectrophotometer (Polaris, Belo Horizonte, MG, Brazil), and results were interpolated from a standard curve, expressed in µmol/L of saliva.

MPO activity as a biomarker of neutrophilic inflammation

MPO assay

Myeloperoxidase (MPO) activity was assessed in duplicate based on the protocol by Gonzaga et al.[8],. Thawed saliva samples were homogenized in a buffer containing hexadecyltrimethylammonium bromide (HTAB, Sigma®) at a 1:20 (v/v) ratio. The homogenates underwent 5 min of sonication, followed by two cycles of freezing at −20 °C for 2 h and thawing at room temperature. Samples were then centrifuged at 5,000 rpm for 20 min at 4

°C and frozen again for 24 h. Afterward, 7 µL of the supernatant and 200 µL of a reading solution (composed of distilled water, potassium phosphate buffer, O-dianisidine, and 1% hydrogen peroxide) were added to a 96-well plate. Absorbance was measured at 450 nm using a spectrophotometer (Polaris®) at 0 and 1 min. MPO activity was expressed as units per milliliter of saliva (U/mL).

Antioxidant assays

SOD estimation

Superoxide dismutase (SOD) activity was determined in duplicate using the method described by Mármora et al.[9], with modifications. Saliva samples were homogenized in phosphate buffer (0.4 M, pH 7) at a 1:10 (v/v) ratio on a refrigerated surface, then centrifuged at 10,000 rpm for 15 min at 4 °C. A mixture of methionine (0.25 mL), riboflavin (0.03 mL), and nitroblue tetrazolium (0.01 mL) was added to 0.1 mL of the supernatant in a 96-well plate. The plate was then exposed to 15 W fluorescent light in an illumination chamber for 10 min. Following exposure, optical densities were immediately measured at 492 nm using a spectrophotometer (Polaris®). SOD activity was expressed in units per milligram (U/mg).

GSH levels

Glutathione (GSH) levels were assessed in duplicate following the protocol by Mármora et al.[9], with adaptations. Samples were homogenized in EDTA (0.02 M) at a 1:10 (v/v) ratio on a refrigerated surface. The homogenate was diluted 1:1 with a solvent composed of 80% distilled water and 20% trichloroacetic acid and centrifuged at 3,000 rpm for 15 min at 4 °C. To 0.1 mL of the resulting supernatant, 0.2 mL of TRIS buffer (0.4 M, pH 8.9) and 0.025 mL of DTNB (5,5'-dithiobis-(2-nitrobenzoic acid)) were added in a 96-well plate. Absorbance was read at 420 nm using a spectrophotometer (Polaris®), and results were interpolated using a standard curve, expressed in µmol/L of saliva.

Inflammatory cytokine evaluation (IL-6 and IL-10)

Levels of interleukin-6 (IL-6) and interleukin-10 (IL-10) were quantified using commercial ELISA kits (R&D Systems, Minneapolis, MN), according to the manufacturer's protocol as referenced by Mármora et al.[9] Cytokine concentrations were reported in pg/mL. The detection range was 62.5–4000 pg/mL, with a lower detection limit of 12.5 pg/mL.

Statistical analysis

Statistical analyses were performed using GraphPad Prism software (GraphPad Software, San Diego, CA). Within-patient time trends were analyzed using the Friedman test with Dunn´s post hoc test to evaluate changes across pre-, mid-, and final-RT. Independent comparisons between the control group (healthy individuals) and the pre-, mid-, and final-RT time points in both protocols were analyzed using the Mann–Whitney test. Comparisons between two groups were performed using the Mann-Whitney test. Chi-square or Fisher's exact tests were used to assess categorical variables such as adverse effects (OM). The significance level was set at $\alpha = 0.05$ ($p < 0.05$). Descriptive statistics were presented as relative and absolute frequencies.

Data availability

The datasets generated during the current study are available from the corresponding author on reasonable request.

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Declarations

Competing interests

M.D.M. would like to disclose that she serves at Azena Medical as a consultant. The other authors declare no competing interests.

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